

SAMUEL HERNANDEZ BALDERAS

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EDUCATION

College of Staten Island, Staten Island, NY

Expected Graduation, May 2027

Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 3.8/4.0

Relevant Coursework: Data Structures & Algorithms, Database Systems, Computer Architecture, Networking and Security, Operating Systems, Digital Systems Design, Web Development, Mobile App Development

SKILLS

Languages: Python, TypeScript, Java, JavaScript, C++, SQL

Frameworks & Tools: Next.js, React.js, CDK, Cloud Formation, Model Context Protocol (MCP) Servers, Git

Cloud & Data: S3, SNS, SQS, ECS, Lambda, Glue, Athena, CloudWatch, API Gateway, Apache Iceberg, Parquet

EXPERIENCE

Amazon, New York, NY

June 2026 – August 2026

Software Development Engineer Intern

- Designed and deployed 3 read-only APIs on an Apache Iceberg/Parquet data lake with paginated, snapshot-consistent cursors and column projections, reducing end-to-end query latency from 52s to under 2s and per-query I/O by up to 60% across datasets with 200K+ rows through tiered caching, file-pruning, and sort-compaction optimizations serving granular billing breakdowns to enterprise customers.
- Developed a Cloudscape demo dashboard backed by a serverless API Gateway + Lambda proxy with SigV4 authentication, enabling stakeholders to validate billing breakdowns across all 3 API surfaces end-to-end.
- Owned the feature end-to-end from data modeling through production deployment: API design, Iceberg read engine, CDK infrastructure, CloudWatch observability, and comprehensive hand-off documentation.

Amazon, New York, NY

June 2025 – August 2025

Software Development Engineer Intern

- Developed an AWS bill computation analytics platform using Python to automate credential-based data retrieval and Athena query execution, building an S3-backed reusable query library via CLI flags, eliminating repetitive manual log searches across environments and reducing ticket resolution time by 83%.
- Designed an end-to-end automated billing data pipeline integrating S3 and Athena with dynamic AWS credential rotation, enabling seamless query execution at scale and reducing analysis and query time by 33%.
- Engineered an MCP Server in TypeScript with Zod schema validation, unifying multiple billing tools into a single AI-powered interface and enabling cross-org accessibility, reducing context switching overhead by 85%.

Amazon, New York, NY

June 2024 – August 2024

Amazon Future Engineer Intern

- Leveraged AWS services (S3, SNS, SQS, Lambda, Glue) to enhance the BillingEgress team's invoicing process, designing scalable data flows that reduced latency and streamlined data processing time by 30%.
- Implemented scalable infra by provisioning AWS CloudFormation resources via Python CDK, focusing on modular design by isolating IAM roles and service-specific configurations across multiple directories.
- Prototyped and transitioned AWS services from a manual Management Console to an automated CI/CD deployment framework using Python CDK and IntelliJ improving system maintainability by 40%.

PROJECTS

Interleave – Explicit-State Model Checker | [Github](#) | Java, Gradle, JUnit, PIT

- Built a Java explicit-state model checker for concurrent shared-memory programs, enumerating thread interleavings and generating replayable traces; detected all 6 seeded concurrency bugs across a 432-test suite.
- Implemented static and dynamic Partial-Order Reduction (DPOR) with happens-before race detection, cutting explored states by 74%, and applied a Bloom-filter bitstate store to reduce the search space to a DAG.

PayCore – Full-Stack Payroll Platform | [Live Demo](#) | [GitHub](#) | Next.js, TypeScript, Supabase, Vercel

- Engineered a Vercel-deployed payroll platform with an 8-table PostgreSQL database, supporting bi-weekly payroll, time-entry approvals, overtime calculations, benefits management, and employee paystub generation.
- Secured payroll operations via Next.js Server Actions, PostgreSQL Row Level Security, and by 243 Vitest tests